

Survey on Cloud Database Security: Cryptographic Techniques, Intrusion Detection and Intelligent Defense Mechanisms

Rachana V. Mahule¹, Sandeep M. Jadhav²

^{1,2}P. R. Pote Patil College of Engineering and Management, Amravati, Maharashtra, India
rvmahule@prpotepatilengg.ac.in, smjadhav@prpotepatilengg.ac.in

Abstract- The fast development of cloud computing drastically changes the way data is stored and managed and has brought about new security problems. The cloud databases are the backbone of enterprise/ mission-critical apps and also the most vulnerable to threats like Unauthorized access, Data leaks, Insider attacks and Advanced persistent threats. In this paper, we conduct a survey of secure cloud database techniques under three aspects: cryptographic-oriented, intrusion detection system (IDS) and intelligent defense. We analyze Crypto-based Solutions (HE, ABE & blockchain based approach) and demonstrate their applicability to address the formulated problem. We also show what machine learning- and deep learning-based IDS can do for instant detection of advanced attack patterns. Finally, we consider a proactive security building mechanisms including anomaly detection, access control and privacy preserving models. According to the research trend analysis made before, this survey gives future challenges and opportunities for any strong and intelligent cloud database security mechanism.

Keywords: Cloud Database Security, Cryptographic Techniques, Intrusion Detection, Intelligent Defense Mechanisms, Privacy Preservation

1. Introduction

Cloud computing is coming fast to the front as a key paradigm for storing and managing of data in today's world providing to companies an edge on scalability, flexibility and infrastructure cost cuts. As more and more enterprise applications rely on cloud databases, security of both the stored data and processed data has become challenging. Cloud environment exposed databases security situation, which opened to an unauthorized person access, endured the insider attacks and data misappropriation concern, or was attacked by the

advanced persistent threats (APTs) [1]. Old-style security techniques, including access control and old-style encryption are not strong enough against the sophisticated threats on cloud paradigms [2]. To address the above issues, latest cryptographic techniques such homomorphic encryption [3], attribute-based encryption and searchable encryption are proposed to realize light-weight secure data processing in confidentiality [4]. On the other hand, IDS is required to watch user behaviour and identify anomalous activities. To detect abnormal pattern activity, machine learning and deep learning models have recently been applied in cloud database that provided real-time protection [5].

Encryption and IDS isn't the only way to enhance cyber-resilience - smart defence is the new buzzword. To address such a challenge, we propose adaptive access control and anomaly-based detection as well as federated learning scheme in order to realize the dynamic, distributed with privacy preserving defense mechanism [6]. These developments illustrate the need for a wide-ranging survey on cloud DB security, especially in incorporating crypto schemes, ID and AI-defense techniques integrated.

We provide an overview of the current status of the art and recent developments in this review with respect to constraints, advantages, open issues and perspectives. By organizing the above cryptographic methods, IDS approaches and intelligent security schemes, etc., this paper is expecting to supply an overall view arranged with cloud database for academic research scientists as well as application researchers.

2. Cloud Database Security Landscape

Cloud databases are the current focus as cloud offers scalability, elasticity and new security pitfalls to have a company to invest in infrastructure. Cloud platforms differ from traditional databases, in the sense, that their backends are distributed multi-tenancy systems where multiple customers share computation and storage. Such architecture also increases the attack surface and it is hard to really achieve data isolation, insiders abuse, regulation compliance [7].

2.1 Architecture of Cloud Databases

There are various cloud database systems in existence: public Cloudbased databases, private Cloud database deployments and hybrid deployments including security concerns of them. This data is also deposited on shared infrastructure of public clouds and this can be accessed by other tenants as it is stored in a multi-tenant architecture of cloud, so the threats like: side channel attacks along with data loss becomes more threatening in this scenario. Hybrid architectures strike the balance between security and scalability, while at the same time increasing complexity for trust boundaries management [8].

2.2 Threat Models in Cloud Environments

Cloud databases face a wide range of threats, including:

- **Insider Attacks:** Authorized users exploiting privileged access to misuse or leak sensitive data.
- **External Threats:** Malware injections, distributed denial-of-service (DDoS) attacks, and SQL injections aimed at database services.
- **Advanced Persistent Threats (APTs):** Sophisticated adversaries maintaining long-term access to databases through stealthy techniques [9].

These threats highlight the necessity of combining preventive and detective measures, as traditional access control alone cannot ensure complete protection.

2.3 Key Security Requirements

The fundamental requirements for securing cloud databases are often summarized as:

- **Confidentiality:** Preventing the unauthorized distribution of sensitive information. This requirement is fulfilled by cryptographic techniques such as homomorphic and attribute-based encryption [10].
- **Integrity:** Ensuring data is accurate, complete and cannot be tampered with. More and more, blockchain and digital signatures are combined to ensure data integrity [11].
- **Availability:** Securing databases from service disruptions and maintaining continuous access during malicious attacks, such as those perpetrated by DDoS. Fault tolerant IDS and surviving architectures are very important at this point [12].
- **Accountability:** Enabling traceability and auditability of database transactions, often implemented through logging, monitoring, and forensic techniques [13].

Overall, the security landscape of cloud databases demonstrates that cryptography, IDS, and intelligent defense mechanisms must operate in synergy to ensure end-to-end protection in distributed environments.

3. Cryptographic Techniques for Cloud Database Security

Cryptography is at the heart of securing cloud databases, which constitutes the underlying technology to ensure data privacy, data integrity and secure computation on sensitive information. With increased prevalence of cloud deployment maximum, basic encryption algorithms fail to keep up with current security requirements and embedded/application layer level ciphers are replaced by high order ciphering techniques [14].

3.1 Symmetric and Asymmetric Encryption

Challenge with Secure Data Storage: When cloud requires to store bulk data securely, symmetric encryption algorithm such as AES is commonly used. The Testing Stack Symmetric solutions can not, however, perform as a shared key management for the users across a network. The asymmetric schemes of RSA, ECC also exhibit inherent scalability in authentication, secure key establishment and exposure resistance but at a high computational cost [15].

3.2 Homomorphic Encryption (HE)

HE compute on encrypted data and does not require the data to be decrypted so is a good fit for processing sensitive data on untrusted cloud servers. A second limitation is that Fully Homomorphic Encryption (FHE) which is the most general, but still it has an intensive computation. Partially and somewhat HE schemes have pragmatic tradeoffs to be applied on real systems [16]. The recent works focus on achieving efficiency using lattice based and ring learning-with-errors (RLWE) cryptosystems [17].

3.3 Attribute-Based Encryption (ABE)

ABE further strengthens coarse-grained access control by including the decryption key issuing on attributes of users; it is suitable for multitenant cloud systems. It enables expressive policies for sharing the data but has issues with scalability and revocation [18].

3.4 Searchable Encryption (SE)

SE search on encrypted data through the cloud confidentially and efficiently. Symmetric SE is the most efficient (leakage on this type of scheme was considered) and public-key ones give better security, at a higher computational cost [19].

3.5 Blockchain-Enabled Security Frameworks

Blockchain is the technology which increases transparency, immutability and trust in cloud databases. Blockchain-based systems achieve secure data provenance and anti-tampering by combining smart contracts with cryptographic protocols [20].

3.6 Comparative Analysis of Cryptographic Methods

Symmetric encryption is fast, but poor at scale in a decentralized system. HE offers a high level of secrecy at the cost of being highly resource-demanding. ABE provides fine-grained access, but adds policy management overhead. SE enhances usability with the threat of leakage, while blockchain strengthens integrity with performance concerns. Hybrid schemes, consisting of a mixture of these methods, are beginning to be developed for more robust security [21].

4. Intrusion Detection in Cloud Databases

IDS are essential to detect the malicious activities, in turn strengthen the security of cloud databases against latest cyber threats. For cloud-based IDS solutions, the challenges are scalability, multi-tenancy and dynamic workloads; a traditional system does not deal with them sufficiently [22].

4.1 Role of IDS in Cloud Database Security

IDS are the second line of defense, makes them observe both attitudes and actions of users as well as traffic for unusual activities that could be signifying for unauthorized access or being attacked. In cloud data storage, IDS can supplement cryptographic defense mechanisms with the real-time monitoring of security situations and support forensic analysis [23].

4.2 Traditional IDS Approaches

Traditional IDS techniques are broadly classified as:

- **Signature-Based IDS:** Detect known threats using predefined patterns or attack signatures. Although effective against familiar attacks, they fail against zero-day exploits.
- **Anomaly-Based IDS:** Build behavioral baselines to detect deviations. These methods identify novel attacks but suffer from high false-positive rates [24].

4.3 Machine Learning-Based IDS

Machine learning has transformed IDS by enabling automated classification of normal and malicious activities.

- **Supervised Learning:** Algorithms such as Support Vector Machines (SVM), Random Forests, and Decision Trees have been applied to datasets like KDD'99 and UNSW-NB15 to achieve higher detection accuracy [25].
- **Unsupervised and Semi-Supervised Learning:** These approaches detect anomalies without extensive labeled data, which is crucial in dynamic cloud environments [26].

4.4 Deep Learning-Based IDS

Deep learning provides better solutions for modeling the sophisticated attack patterns. Convolutional Neural Network (CNN), Recurrent Neural Network (RNN) and Long short term memory (LSTM) models have shown better accuracy in IDSs. These hybrid deep-learning models that combine CNN with RNN/LSTM improve both temporal and spatial assessment of vulnerabilities in cloud databases [27].

4.5 Challenges in Cloud-Based IDS Deployment

Despite the progress in IDSs for clouds, several challenges must be addressed, including imbalance among datasets, false- alarm rates and real-time detection latency as well as increasing demand on distributed nodes. Federated and interpretative I D S, which appropriately tradeoff detection accuracy with transparency are the needed solution to address these limitations [28].

5. Intelligent Defense Mechanisms

Although cryptography and intrusion detection are the base of cloud database security, we do believe that due to cloud's dynamic and distributed nature required intelligent defense systems which should be able to proactively evolve themselves in real time by detecting threat. These techniques integrate AI, anomaly detection and adaptive policies to improve resilience [29].

5.1 Adaptive Access Control Models

Classic Role-Based Access Control (RBAC) is not suitable in dynamic clouds. Adaptive models, such as Attribute-Based Access Control (ABAC) [30] and Context-Aware Access Control (CAAC), utilize attributes of users, environmental situations, and historical activities to enforce fine-grained authorization in real time.

5.2 Anomaly and Behavior-Based Detection

Smart anomaly detection programs study how a user's behavior and transaction patterns to determine normal activities. Fuzzy logic and machine learning are widely employed to reduce false alarm rates and identify insider threats [31].

5.3 Privacy-Preserving Models

Privacy issues are important especially for multi-tenant databases which involve sensitive information processing. Approaches like differential privacy, secure multi-party computation and federated learning are used to keep sensitive information safe also during collaborative security analysis [32].

5.4 Distributed Cloud Security over Federated Learning

We can use federated learning to jointly learn IDS models and/or defense models among nodes while the raw data never be transmitted then. This guarantees privacy preserving and improves the detection accuracy in distributed cloud environments [33].

5.5 AI and Expert Systems to Enable Proactive Defense

Predictive analytics can also make use of advancements in AI/ES, making it possible to identify potential attack vectors before they become a threat. Reinforcement learning based defense agents autonomously modify defense (such as firewall, access rules) configurations in order to cope with instantaneous threats [34].

5.6 Comparison of Defense Mechanisms

Each security defense point responds to targeted security from different attack dimensions: adaptive access control is use for authorization, anomaly detection is used in behavior threat and federate learning improve distributed intelligence. However, the consolidation of those mechanisms in holistic scalable frameworks is still an issue [35].

6. Comparative Review of Techniques

This section summarizes the pros, cons and trade-offs in deploying solutions in the three pillars of cloud database security cryptography, intrusion detection and intelligent defense with an enablement focus toward practicality in multi-tenant elastic clouds.

6.1 Comparison Criteria

These models can be compared through the security strength, query/processing support, latency/throughput impact, scalability & key/policy management, leakage/attack surface and privacy guarantees, operational complexity [36][37][38].

Table 1: Comparative analysis of existing work

Technique / Family	Security Strength	Query / Processing Support	Latency / Throughput Impact	Scalability & Key/Policy Mgmt.	Leakage / Attack Surface	Privacy Guarantees	Operational Complexity	Typical Fit
Symmetric (AES at-rest/in-transit)	High for storage/transport [14]	No computation over cipher text	Low overhead	Simple keys but rotation/sharing at scale can be hard	Low if keys managed well	Confidentiality only	Low	Bulk storage, backups
Asymmetric (RSA/ECC)	High for exchange/auth [15]	Key exchange, auth only	Moderate	Easier distribution, harder per-user ops at scale	Low	Confidentiality/authn	Low–Moderate	Session setup, KMS
Homomorphic Encryption (HE)	Very high [16], [17]	Arbitrary (FHE) / limited (SHE/PHE)	Very High (improving)	Keys simple; circuit design complex	Side-channels from access patterns (not content)	Strong confidentiality during compute	High (scheming)	Sensitive analytics w/ untrusted compute
Attribute-Based	High; fine-grained	Policy-driven	Moderate–High	Challenging:	Misconfig/policy	Strong, policy-	Moderate–High	Multi-tena

Encryption (ABE)	d [18], [36]	n decrypt	(policy eval)	revocation, updates	inference	scoped access		nt sharing, PHRs
Searchable Encryption (SE)	High content confidentiality	Keyword search over cipher text [19]	Low–Moderate	Manage per-index keys	Leakage risks (access/volume patterns) [37]	Strong vs plaintext; weaker vs leakage abuse	Moderate (indexing)	Encrypted logs/docs, metadata search
Blockchain-assisted DB security	Integrity, provenance [20], [42]	Audit/triggere d checks	Moderate–High (consensus)	Global view; throughput limits	Smart-contract bugs; liveness	Immutability, accountability	Moderate–High	Audit trails, tamper-evident logs
CryptDB-style layered encryption	Good practical balance [43]	Limited SQL on cipher text	Low–Moderate	Per-column/key mgmt	Some leakage by design	Practical confidentiality	Moderate	OLTP with limited ops
Classic (Signature/Anomaly) IDS	Detects known/unknown [24], [25]	N/A (monitoring)	Moderate (inline/offline)	Horizontal scale of sensors	FP/FN trade-offs	N/A	Moderate	Baseline monitoring
ML/DL IDS (CNN/RNN/LSTM/Hybrid)	High detection on modern sets [27], [28]	N/A	Moderate–High (training/inference)	Requires data pipelines	Dataset shift; adversarial ML	N/A	Moderate–High	High-throughput cloud traffic
Federated/DP-aware Defense	Strong privacy for training	N/A	Moderate (agg. rounds)	Good for multi-tenant	Poisoning/Byzantine risks	DP guarantees (ϵ,δ)	High (orchestration)	Cross-tenant

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- For confidential analytics on sensitive tables, HE offers the strongest confidentiality at compute time but remains costly; SE gives practical search with known leakage trade-offs [17], [19], [37].
- ABE is attractive for fine-grained, multi-tenant access, but revocation/policy churn is a pain point at scale [18], [36].
- Blockchain improves integrity, provenance, and audit, but throughput/latency constraints limit primary-store use; it works best as an audit/compliance layer [20], [42], [43].
- DL-based IDS consistently outperforms classic baselines on modern datasets (e.g., UNSW-NB15, CIC-IDS2017) but needs careful handling of drift, imbalance, and adversarial examples [27], [28], [40].
- Federated + DP defenses enable cross-tenant learning without raw data sharing, suitable for regulated settings; secure aggregation mitigates information leakage but adds orchestration complexity [33], [38], [39].

6.2 Strengths and Limitations (Narrative Synthesis)

- **Best-in-class confidentiality:** FHE for compute-on-ciphertext; ABE for who-can-decrypt policies; SE for practical encrypted search often combined with KMS, HSMs, and access pattern defenses (ORAM/Oblivious indexing) to curb leakage [16]–[19], [37].
- **Detect-and-respond:** DL IDS (CNN/LSTM/Hybrids) achieves strong detection on UNSW-NB15/CIC-IDS2017, but real deployments need online learning, explainability, and cost-aware inference to keep latency SLOs [27], [28], [40], [41].
- **Provable privacy at scale:** Differential privacy and secure aggregation strengthen collaborative modeling; careful ϵ -budgeting and robust aggregation (Byzantine-resilient) remain open engineering challenges in multi-cloud contexts [32], [33], [38], [39].
- **System co-design matters:** End-to-end architectures that pair cryptography (for prevention), IDS (for detection), and federated/DP (for privacy-preserving intelligence) deliver the most resilient posture—mirroring lessons from CryptDB-style systems and provenance-anchored (blockchain) audit trails [20], [43].

7. Open Challenges and Future Research Directions

Although significant progress has been made in securing cloud databases through cryptographic protocols, intrusion detection systems, and intelligent defense mechanisms, several open challenges remain. Addressing these challenges is crucial for developing scalable, efficient, and trustworthy solutions.

7.1 Scalability and Performance Trade-offs

Fully Homomorphic Encryption (FHE) and ABE provide strong levels of confidentiality, however they incur high overhead. The high overhead of query execution time and resource cost makes it difficult to be applied in large-scale cloud databases [44]. Research should target lightweight encryption schemes and hardware acceleration (e.g., GPUs, FPGAs) to achieve better balance on security v/s performance [45].

7.2 Real-Time Security vs. Computational Overhead

The deep learning-based IDS models attain the high accuracy at the cost of heavy inference overhead that may contradict with real-time Service-Level Objectives (SLOs). The development of cost sensitive IDS models with on-line/streaming learning is still an open research problem [46].

7.3 Multi-Tenancy and Privacy Concerns

The isolation and prevention of information leaking between tenants in multi-tenant databases is still challenging. Federated learning and differential privacy are high-potential studies, but without yet a solution to secure the aggregation in poisoning/adversarial sense [47].

7.4 Integration of Blockchain and AI in Cloud Security

Blockchain ensures immutability and auditability while AI-based models deliver predictive defense. Nevertheless, the consolidation of these technologies faces issues in relation to throughput bottlenecks, energy efficiency and cross-platform compatibility [48]. In this direction, lightweight consensus algorithms and AI-supported blockchain frameworks could be further explored.

7.5 Towards Explainable and Trustworthy Defense Systems

The lack of explainability has been identified as a fundamental obstacle to adopting AI-enabled IDS and defense systems at scale. Black-box models hurt trust and undermine compliance with data privacy laws like GDPR. Explainable AI (XAI) - Future research needs to give more attention to XAI in intrusion detection and defense, maintaining transparency not at the cost of accuracy [49][50].

8. Conclusion

Cloud database systems are becoming essential for next-generation enterprises in terms of scalability, elasticity and efficiency. But their distributed / multi-tenant nature leaves them open to a wide range of security vulnerabilities including insider abuse, data loss and APTs. According to the contributions of this survey, we have conducted a comprehensive study on cloud-based database security against their three main dimensions: cryptosystems, intrusion detection system (IDS), and intelligent defense mechanism.

Encryption methods (homomorphic, attribute-based or searchable encryption) can provide a high level of confidentiality combined with fine-grain access control; however they pay the price in terms of performance and scalability. Intrusion detection systems, particularly by machine and deep learning, improve the accuracy of detection, but dataset imbalance, adversarial robustness and real-time deployment remain open issues. Meanwhile, the intelligent defense such as adaptive access control, anomaly detection, federated learning and AI-based methodologies show great potential for proactive/distributed/privacy-preserving security.

Our study underscores, nevertheless, that there is no one-size-fit-all way of so far cloud database security. Rather, use of hybrid systems that bring content protection through cryptography along with SCP to IDS or smart defense is the best. In coming, avenues of open research problems with study such as improving the efficiency of cryptographic approaches

and develop new explainable, concernful AI defence techniques and integration of blockchains using an AI driven models. Such challenges will need to be addressed to realize secure, scalable and trusted cloud database systems in future data driven ecosystems.

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